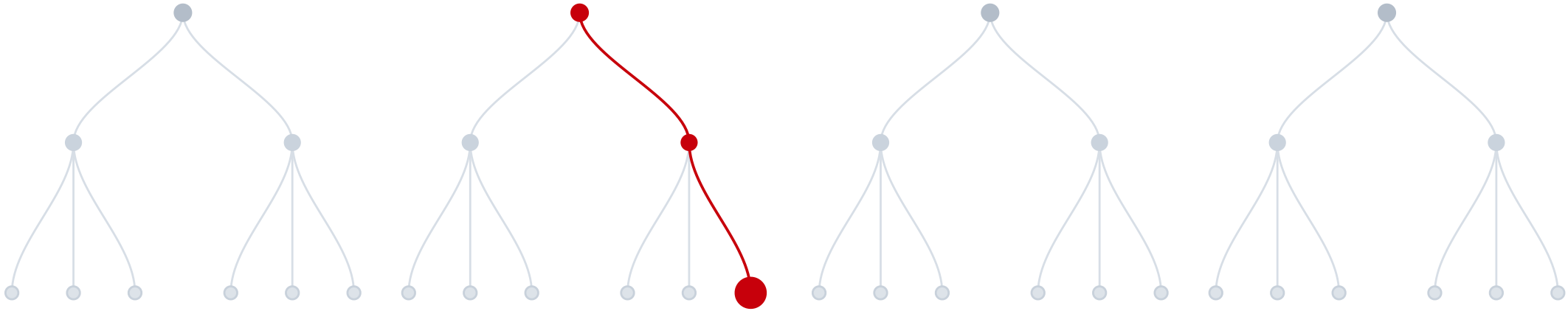
Technical Solution: Hierarchical Discovery Pipeline

A search for a better mathematical solution

In this Erdős example, lower scores are better.



Starting candidate (parent)

Score: 0.509541

Each dot is a saved candidate. Each link records where it came from.

A candidate is a mathematical construction produced by a program.

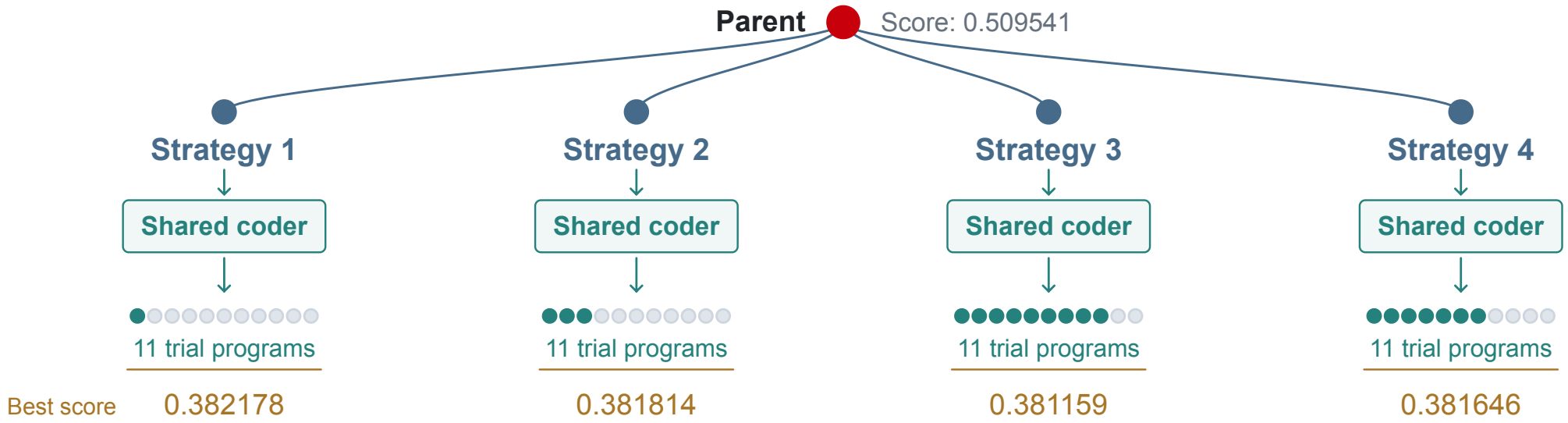
Technical Solution: Hierarchical Discovery Pipeline

Each strategy uses the parent and every earlier strategy



Technical Solution: Hierarchical Discovery Pipeline

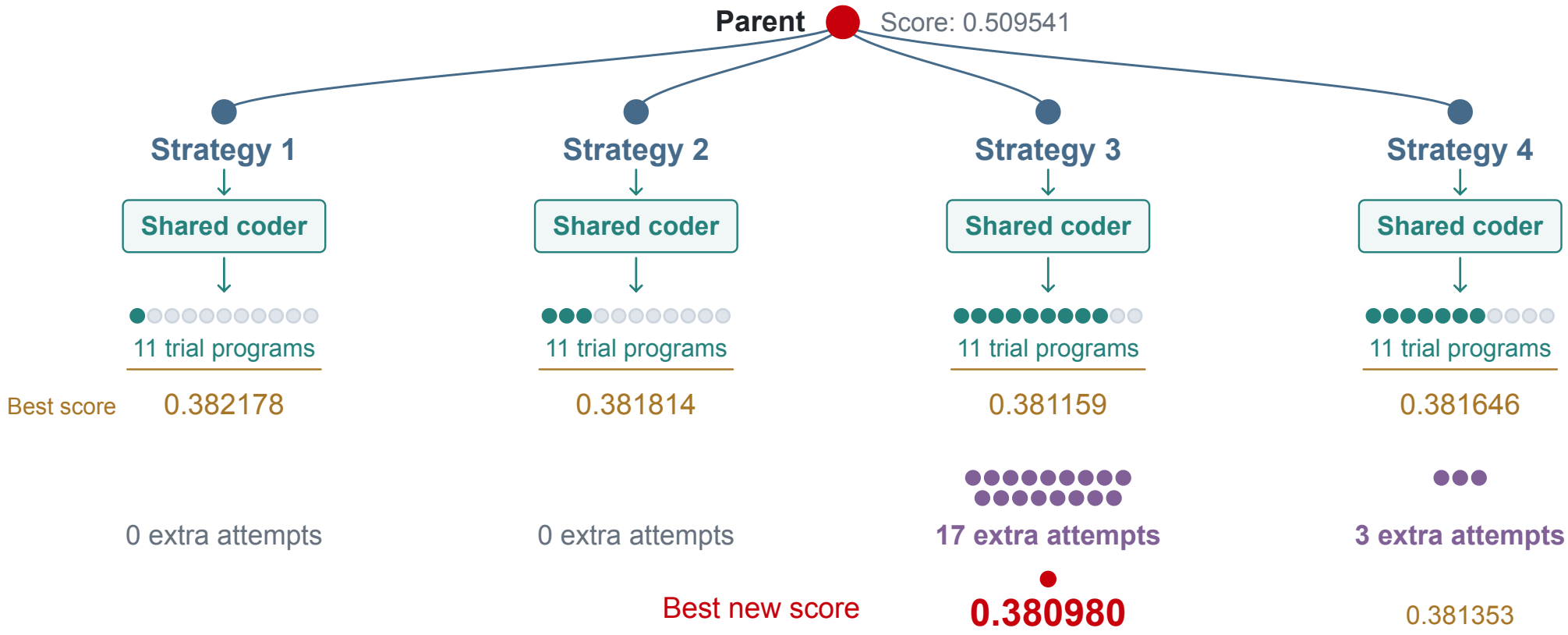
The coder turns each strategy into testable candidates



The evaluator runs the code, checks its output, and assigns a score.

Technical Solution: Hierarchical Discovery Pipeline

The bandit allocates more attempts using pilot results

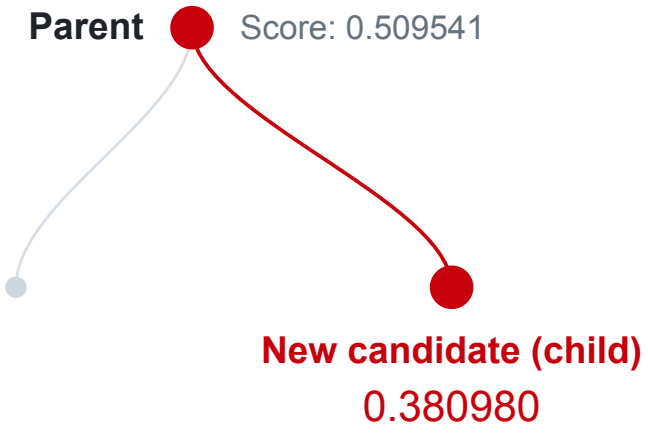


The bandit uses pilot results to assign extra attempts.

Technical Solution: Hierarchical Discovery Pipeline

The system saves candidates and trains the coding model

Saved candidates (archive)



Results from all parents

SPO² training

Coding model

Updated

Learns from evaluated programs

One update after the full batch

Planning model stays unchanged

Technical Solution: Hierarchical Discovery Pipeline

The search repeats with both updates

